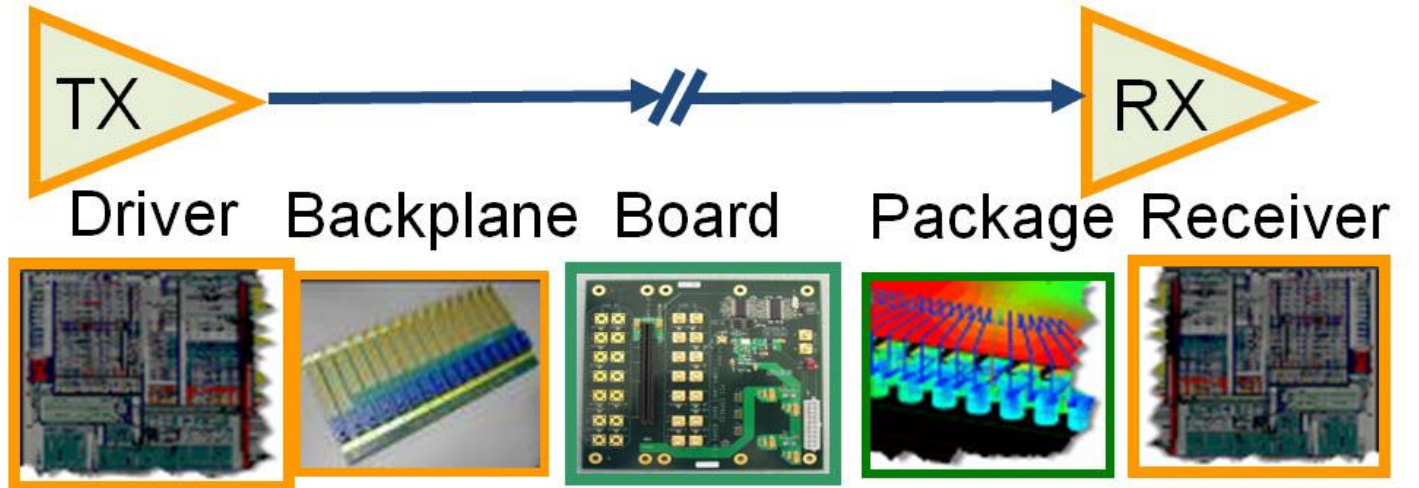


Understanding a Signal Integrity in a PCB



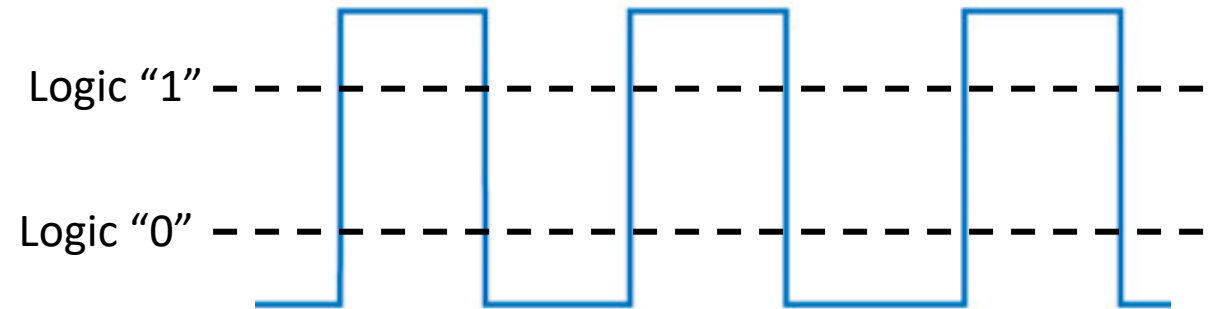
What could possibly go wrong?

◦ By Praveen Sah

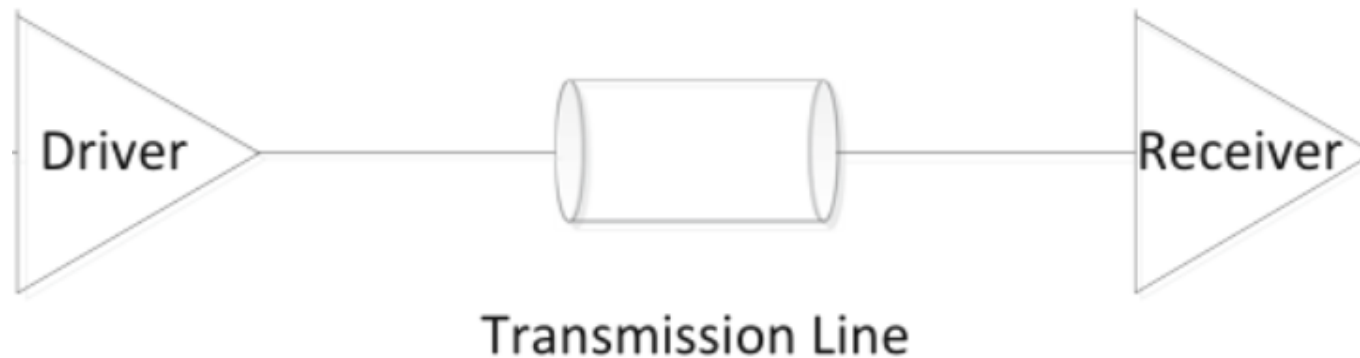
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Signals

- Signals are analog electrical quantities used to transfer information
 - ❖ Often used to represent digital information



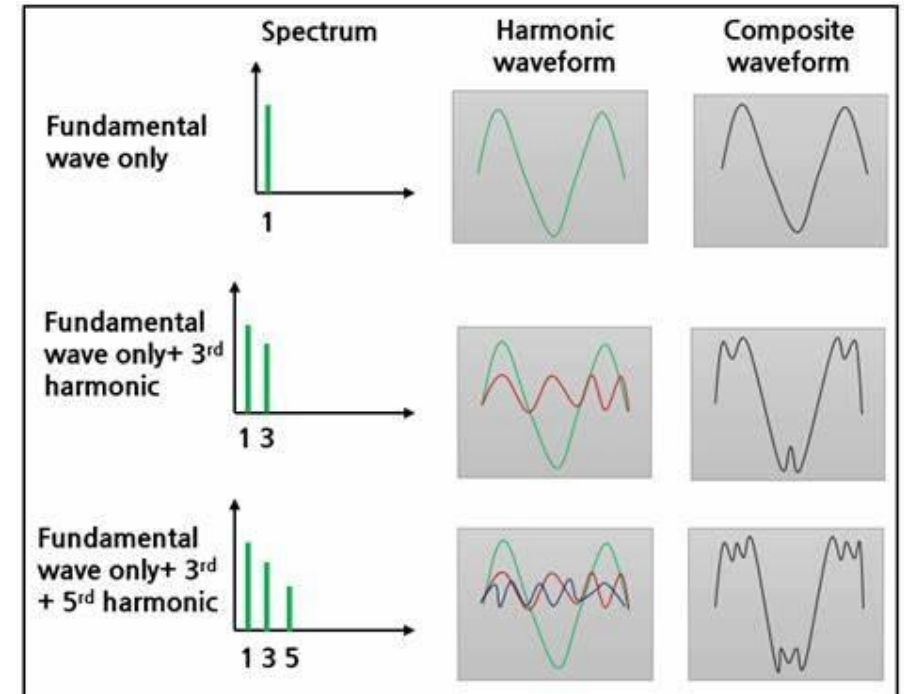
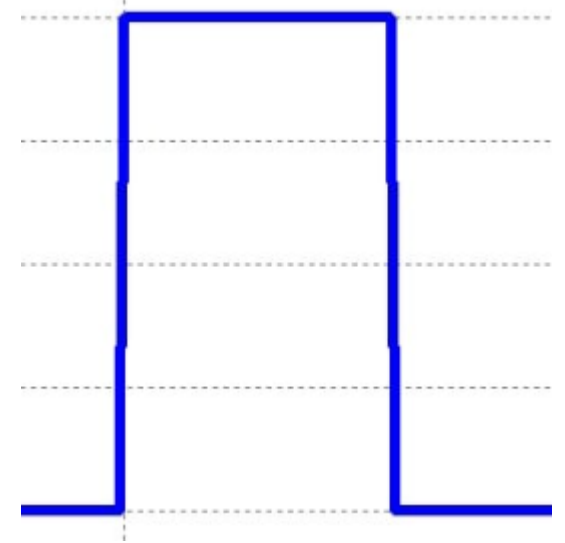
- Signal chain consists of:
 - Transmitter (Driver)
 - Channel (Or Interconnect)
 - Receiver



Requirements for a good data transmission

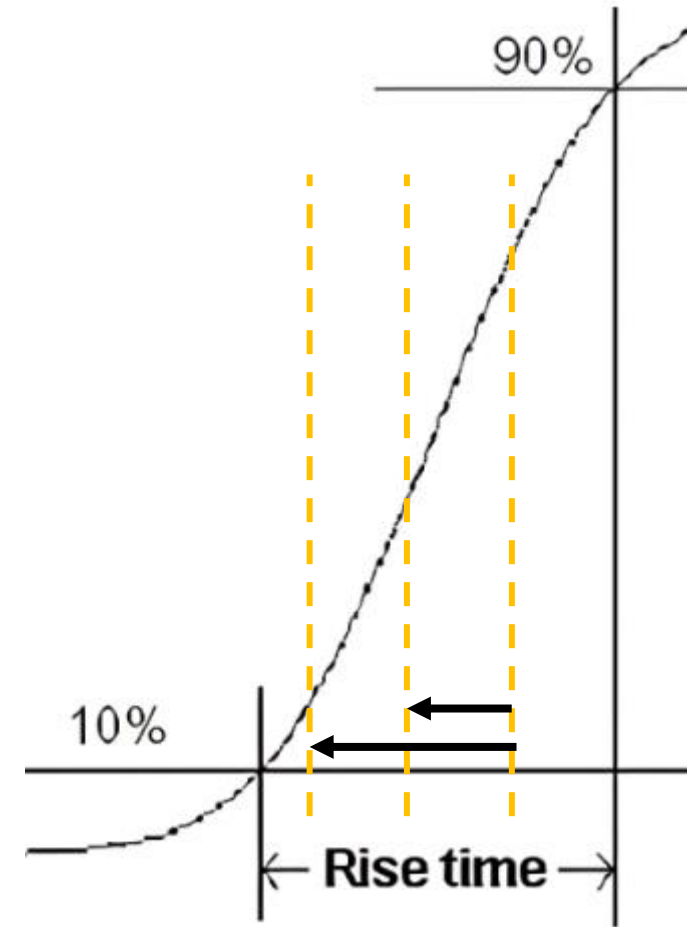
- Most digital data is transmitted as square wave signals
 - consists of multiple frequency components
- Good digital data transmission requires:
 - Constant amplitude change for all frequency components
 - Constant time shift/delay for all frequency components
- If either of these are not constant, signal distortion occur

1. Does the signal reach its destination within the given time limit?
2. When it reaches its destination, is it in good condition?



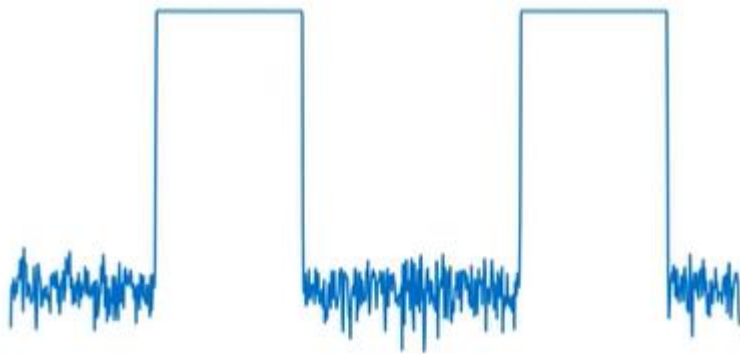
Signal Integrity

- Signal integrity is the ability of a system to transfer (data) signals without excessive distortion
- Signal distortion are present in all systems.
- Earlier we used to ignore because the rate at which signal change was very low or rise time was long .
- But signal integrity becomes important at higher speed (shorter rise times)
- Shorter rise times / higher frequencies (>50MHz). Both digital and analog aspects need to be taken care
- High speed is not strictly defined
 - ❖ Rise times on the order of ~ 1 ns
 - ❖ SI is an important concern in high-speed digital design
- When a signal propagates from the driver to the receiver, it doesn't remain the same, whatever has been sent originally will be received with varying degree of distortion. This signal distortion happens due to factors like impedance mismatch, reflection , crosstalk, jitter, and PCB ringing. A designer aim should be minimizing such factors so that original signal could reach to the destination without excessive distortion.

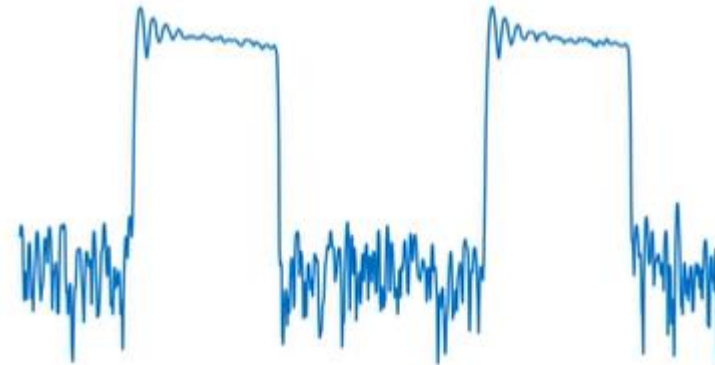


Channel

- Ideal Channel: Received signal has different magnitude or time shift
 - Signal “shape” remain the same
- Real channel : frequency dependent components can change the “shape” of the received signal.
- Usually, high frequency components are attenuated more than low frequency components.



Ideal channel



Real channel

Channel formats

- Channel can have different physical formats
 - Printed circuit board (PCB) traces
 - Connectors /backplane (between boards)
 - Cables (between components/systems)



Why maintaining a signal integrity is a challenge?

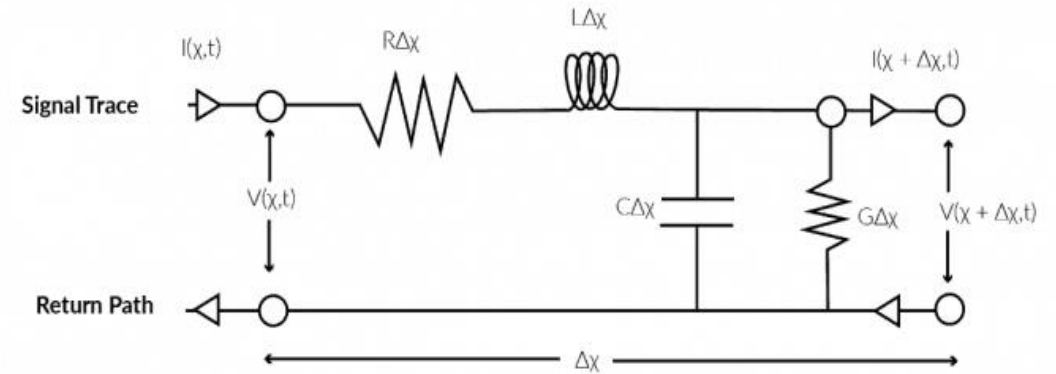
- Technological advancements (high density Interconnects, scaling down of technology node etc.)
- Rising Bandwidth
- Faster Bus cycle
- Shorter Rise time
- Board Space

- Circuit board traces act as transmission lines when driven by signals with shorter rise times (~ns)
- When does traces start to exhibit transmission line behavior? as a transmission line model?
 - The answer is when the trace length exceeds a critical length.

$$Z = R + jX, \quad X = \text{Reactance}$$

- At low frequency, trace resistive characteristics come into play and at higher frequency, the inductance and capacitance is a point of concern
- Maximum frequency in a signal is a property of rise and fall times.

- $F_{\text{max}} \text{ (GHz)} = 0.5 / T_{\text{rise}} \text{ (ns)}$

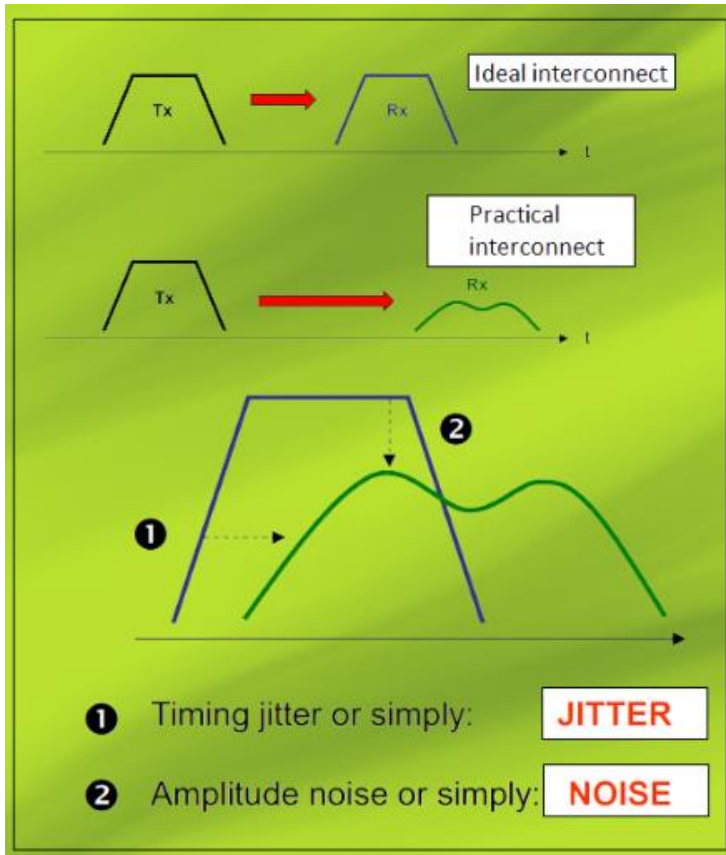


$$L_{crit} = \frac{1}{6} \frac{T_{10-90\%}}{T}$$

Factors affecting Signal Integrity

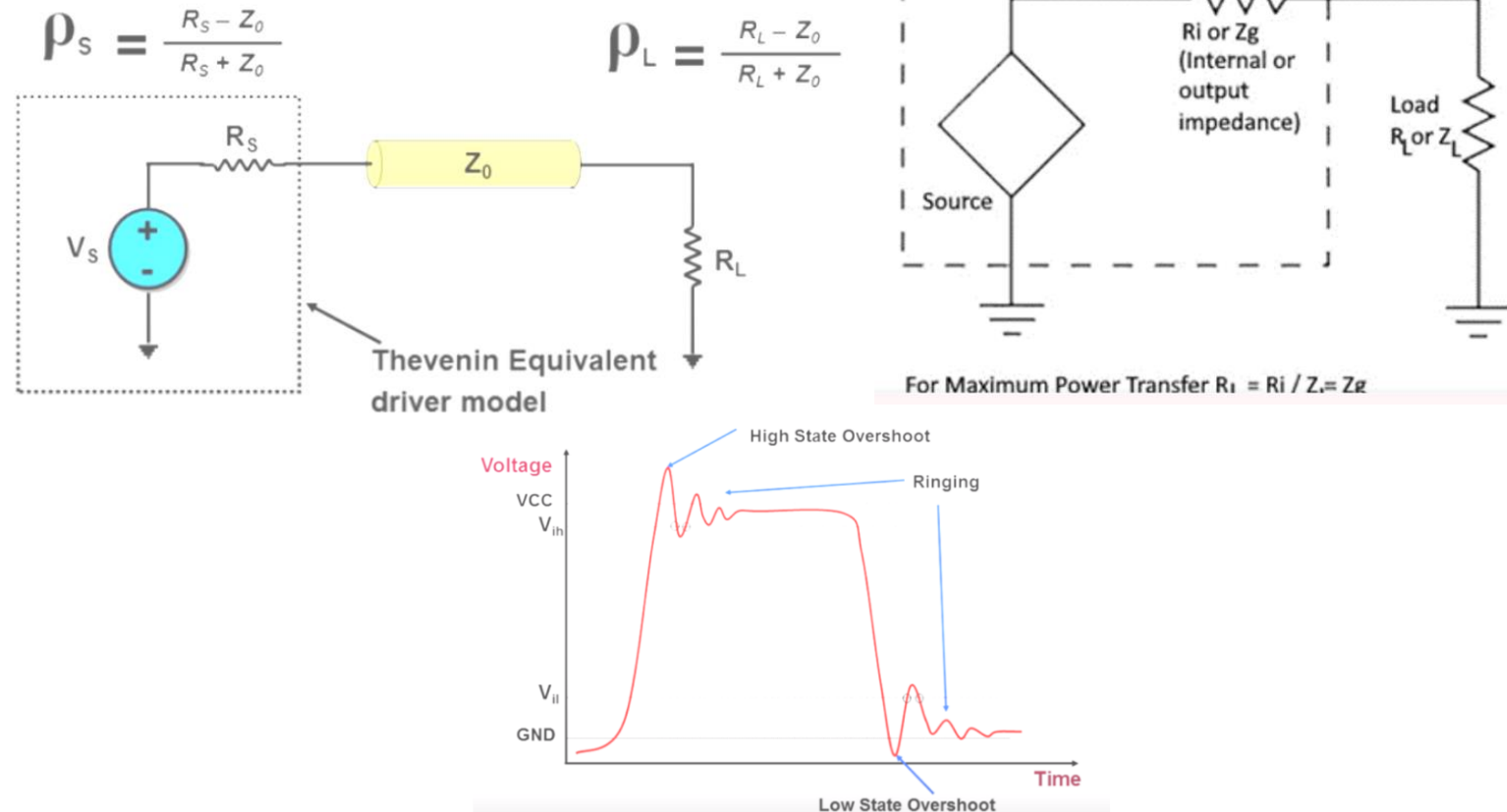
➤ Transmission Line Effects

- Trace width, routing, vias, critical length



➤ Impedance mismatch effects/Reflection

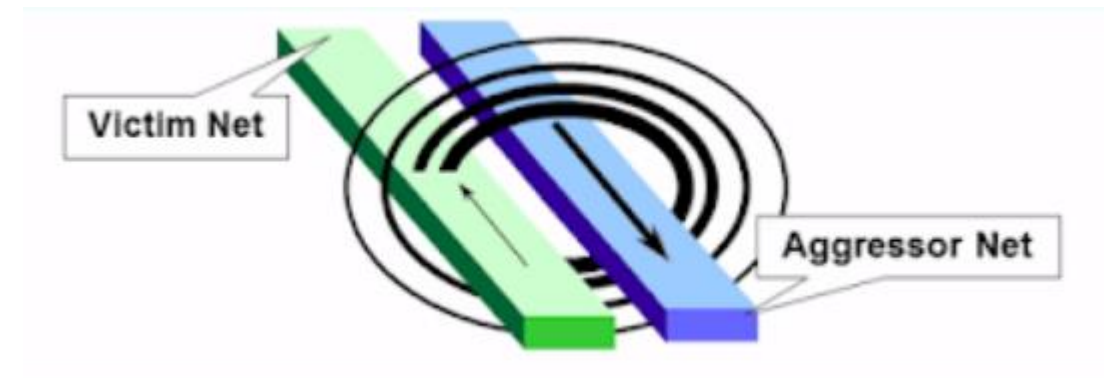
- The load impedance of the output port must match the characteristics impedance or the internal impedance to maximize signal coupling between the two ports.



Factors affecting Signal Integrity

- **Electromagnetic Interference (EMI):** Every switching operation creates a certain amount of noise. These noise can be radiated by traces acting as antenna, the strength of radiation signal is proportion to the switching frequency.
- **Timing (flight time) :** determine the flight times of the signals as they travel through the board. Board timing is an important part of the system timing, and gets affected by the lengths of the lines, their propagation speed as they travel through the board, and the shape of the waveform at the receiver.
- **Crosstalk:** When two nets close together, a change in current flow in one net can introduce current in an adjacent.
 - Aggressor Net: A neighbouring net with changing the current
 - Victim Net: A net at a steady-state

Aggressor net contributes the crosstalk in the victim net when a driver in the aggressor net is switching



How to solve Signal Integrity problems

- ☐ Simulate the channel
- ☐ Identify the root cause of degradation
- ☐ Explore design solutions

- Define requirements: Understand the characteristics and specifications of high-speed board. Identify the key parameters such as data rate, transmission medium, clock skewness

- Modelling components: Break down the board into components – traces, vias, connectors and terminations
 - Create model for each component using appropriate techniques (lumped –element models, transmission line models)

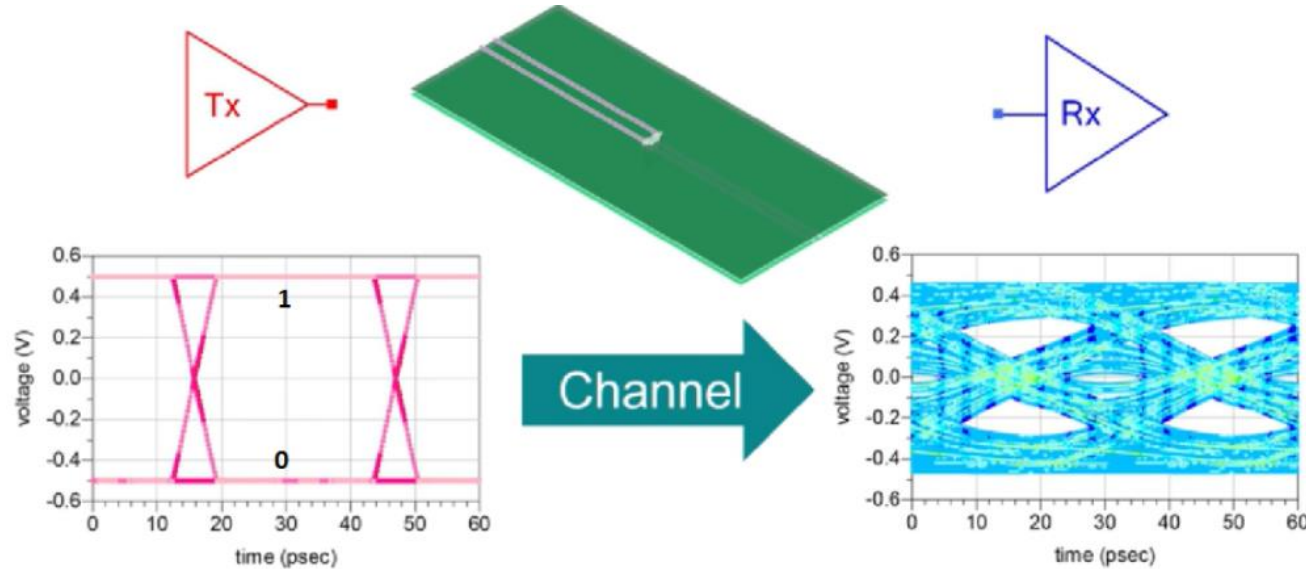
- Transmission line modelling: Implement transmission line models for signal path considering parameters like impedance, propagation, delay and attenuation
 - Account for non-ideal effects such as skin effects or dielectric loss

- Simulate signal propagation
 - Utilize simulation tool such HyperLynx, Siwave to simulate the signal propagation through the channel.
- **By Praveen Sah**

Simulate the channel

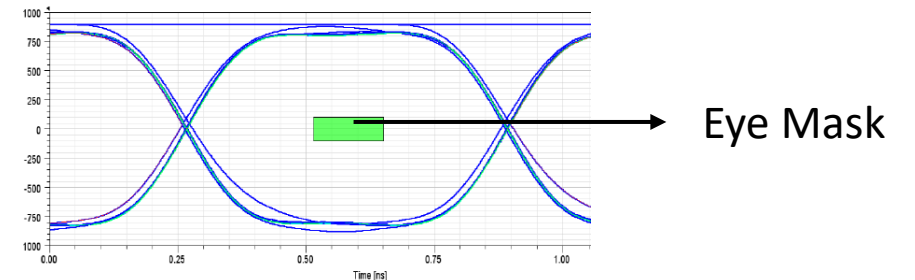
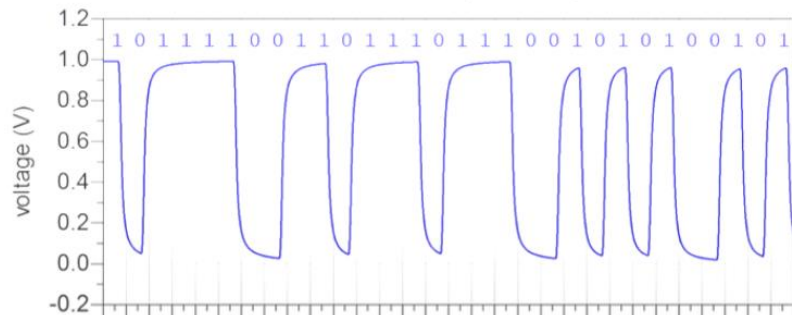
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Eye diagram tell how much channel degrades the transmitted signal



How is eye diagram formed?

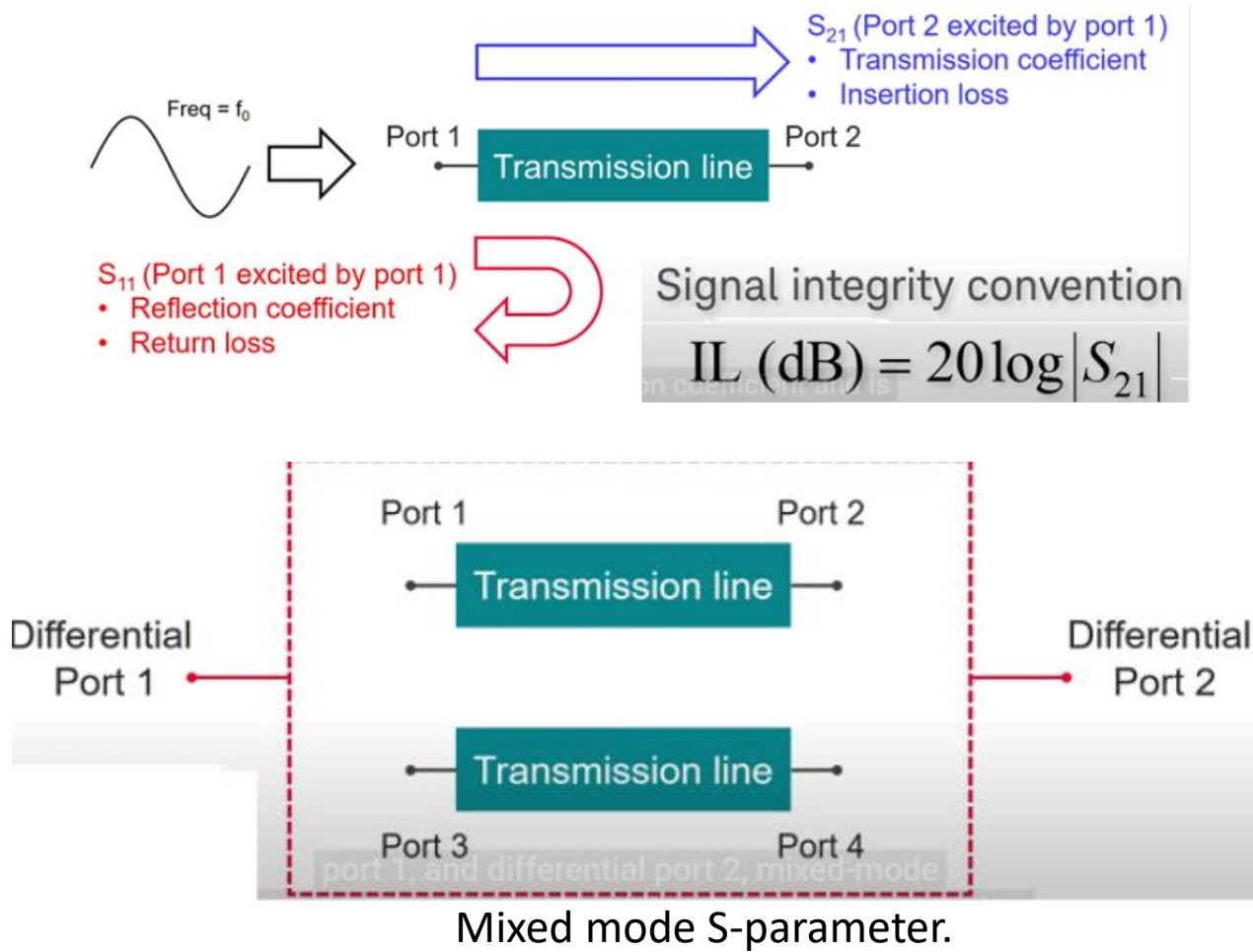
Pseudo-Random Binary Sequence (PRBS)



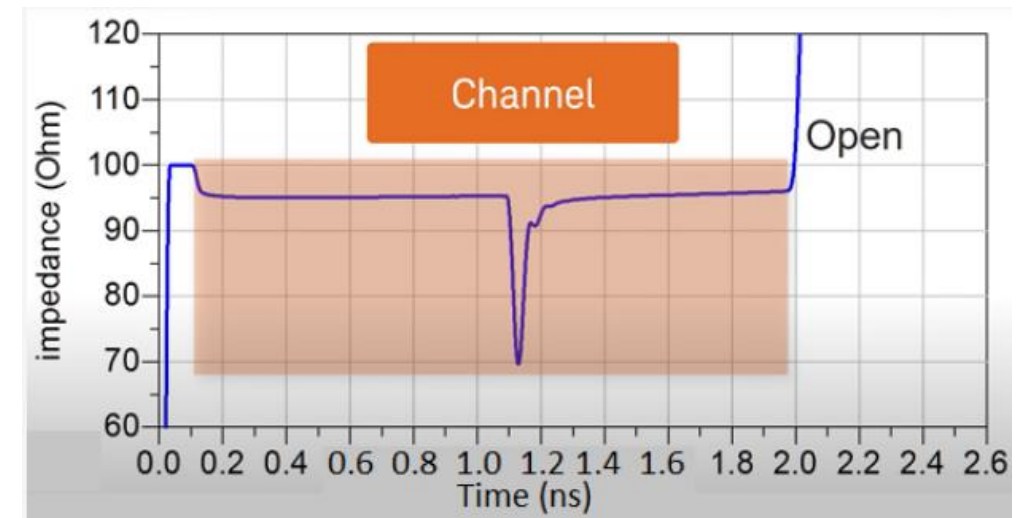
The PRBS is sliced at unit interval and overlayed so that rise and fall time are synchronized

Identify the root cause of degradation

- Use S-parameter analysis and time-domain reflectometry.



S-parameter give the frequency response, we can use time-domain reflectometry to deduce the spatial and timing information. On the TDR plot, the right-hand side showcases the 'open' circuit to help us identify the channel.



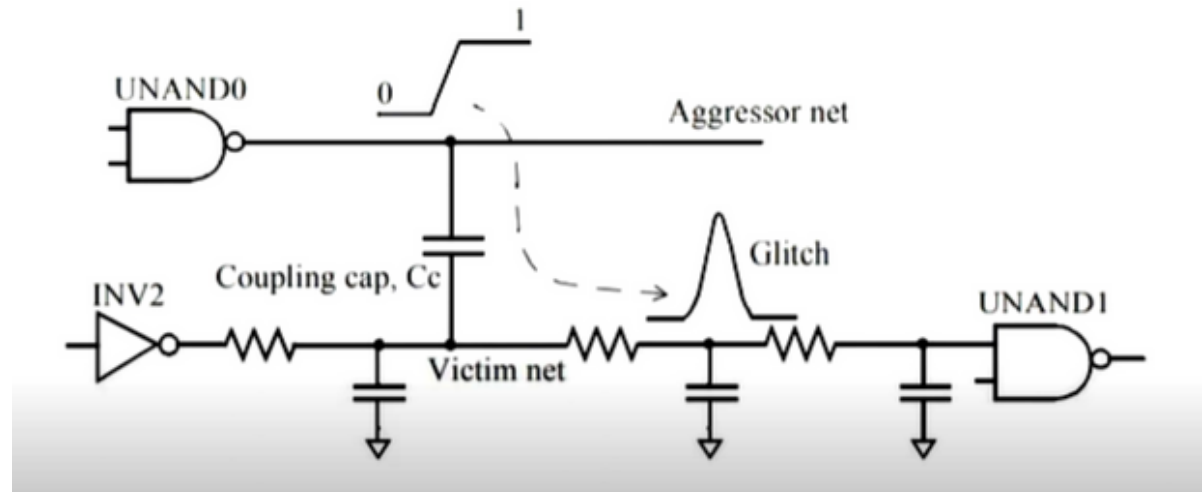
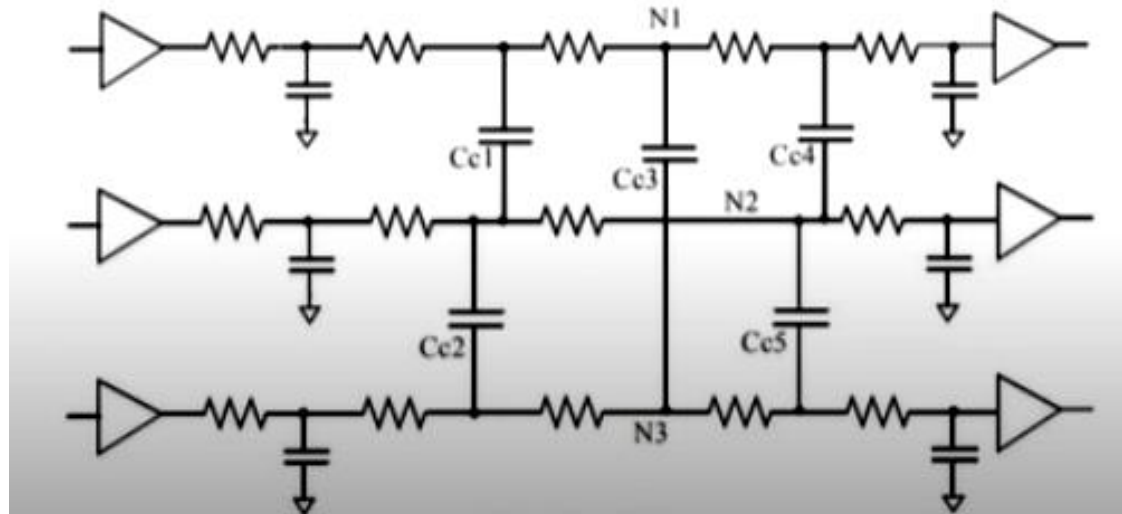
These parameter are used, whenever we use a pair of transmission lines to transmit a differential signal from differential port 1 and 2. This tell us how the transmission linereacts to differential and common mode signals

Explore design solutions

- **Impedance matching:** impedance of source, load and signal traces should be matched Terminating resistors, length of stubs.
- **Minimize the effects of propagation delay:** Signal skew can be minimized by matching the length of signal traces
- **Reduce signal attenuation:** use of low-loss dielectric material and low resistance traces will minimize signal attenuation
- **Reduce crosstalk:** maximize distance between signal traces, consider using differential which are less sensitive to noise, don't split return paths
- **Minimize supply voltage fluctuation:** use of power and ground planes/polygon , decoupling, vias current
- **EMI reduction:** Use a ground plane, minimize loop area, keep trace inductance lower

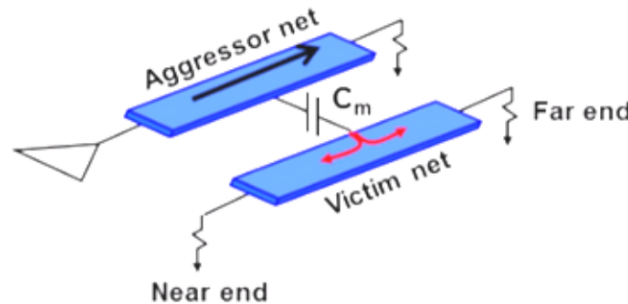
Crosstalk

- Crosstalk is the undesired electrical interaction between two or more physically adjacent due to capacitive cross coupling and Inductive coupling.



Crosstalk due to Electric field

- When crosstalk happens, we say two traces are couple and coupling happens due to the Electric field. The Electric field can be represented by mutual capacitance which injects current to victim proportional to the current change in Aggressor net. There is the mutual capacitance between two nets due to their proximity. This capacitance represents the Electric field that gets created because of changing current in the aggressor net.



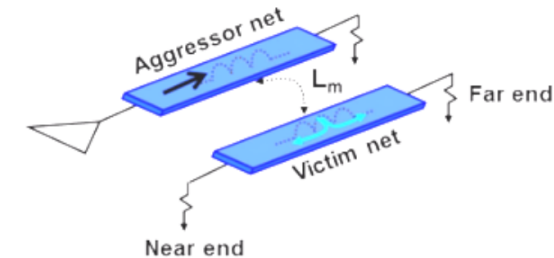
Mutual capacitance injects **current** onto the victim net proportional to the rate of change of voltage in the aggressor net.

$$I_{\text{noise}} = C_m \frac{dV_{\text{aggressor}}}{dt}$$

- The effects of Electric field and mutual capacitance inject energy in the victim net and causes current flow in both directions. These unwanted effect is called crosstalk.
- The crosstalk that happens closer to the aggressor driver called near end and the opposite is far end.

Crosstalk due to Magnetic field

- The coupling between two conductors due to a magnetic field is represented by mutual inductance. Similar to the Electric field, we have the near and far end.



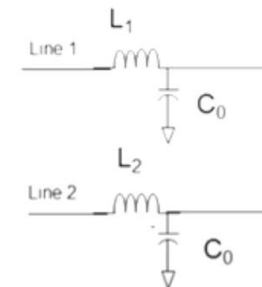
Mutual inductance induces **voltage** onto the victim net proportional to the rate of change of current in the aggressor net.

$$V_{\text{noise}} = L_m \frac{di_{\text{aggressor}}}{dt}$$

Uncoupled and Coupled transmission Lines

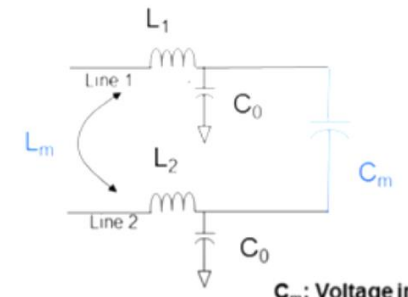
- Uncoupled transmission lines
 - When voltage and current on line1 does affect the current and voltage on line 2
- Coupled transmission lines
 - Energy is coupled from one transmission line into another through Electric and Magnetic which are represented by mutual Capacitance and inductance

Uncoupled transmission Lines



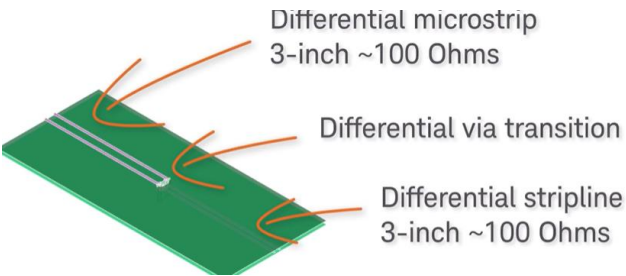
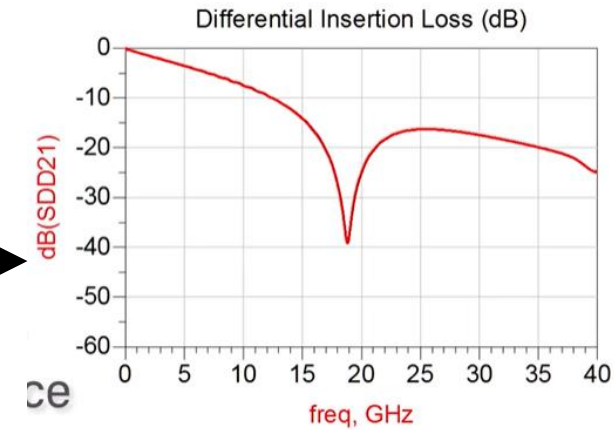
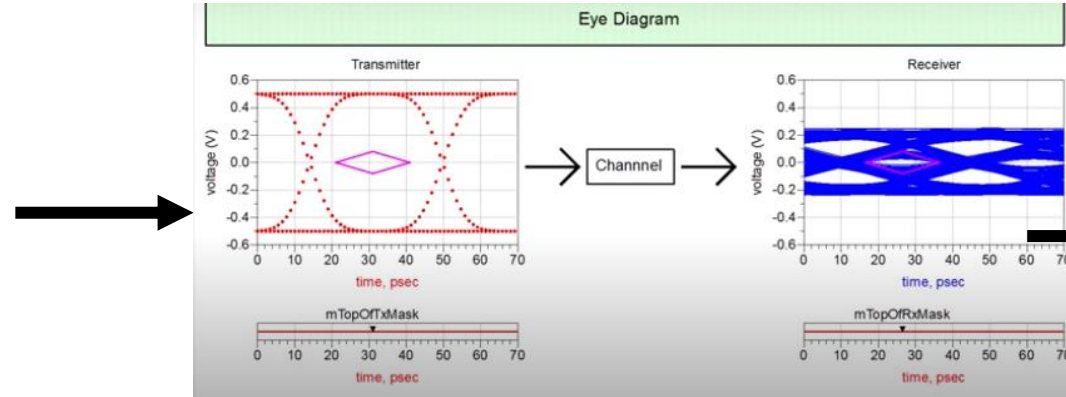
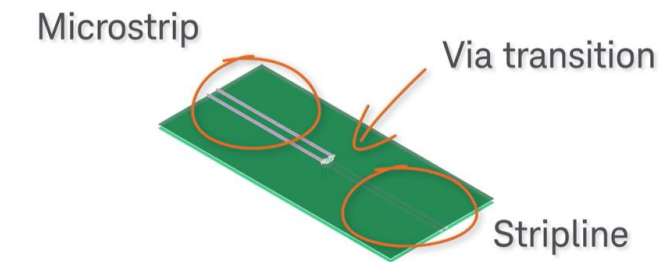
Current and Voltage on line 1 does not affect line 2

Coupled Transmission Lines



C_m : Voltage in line 1 induces current on line 2

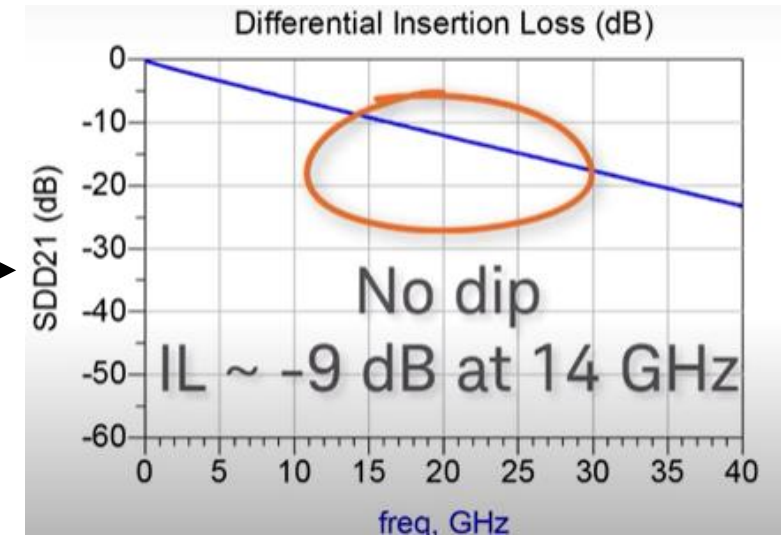
Case Study -1



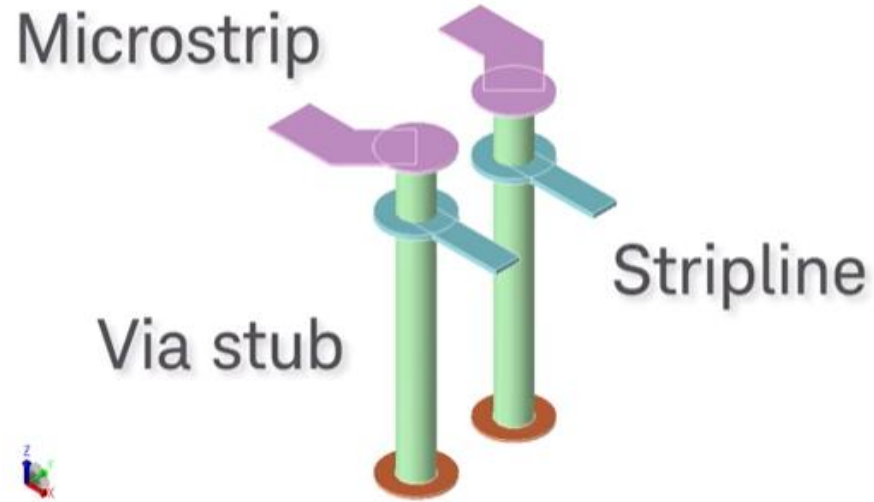
Data rate: 28 Gbps
Nyquist Frequency = data rate/2
Attenuation: ~ 0.1 dB/inch/GHz

$$\frac{0.1 \text{ dB}}{\text{inch GHz}} \times 6 \text{ inch} \times 14 \text{ GHz}$$

= 8.4 dB



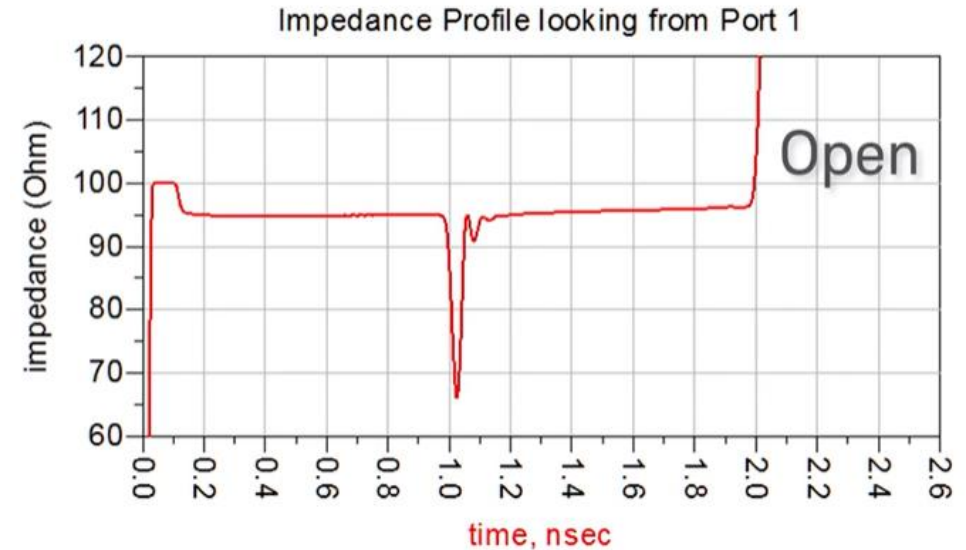
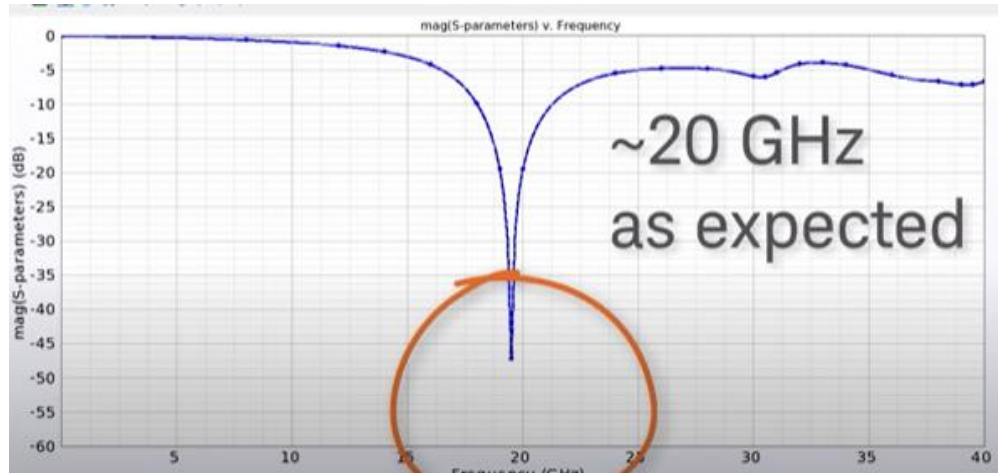
Case Study -1



$$f_{\text{res}} = \frac{1.5}{\text{stub length (inch)}}$$

Stub length: ~75 mils = 0.075 inch

$$\frac{1.5}{0.075} = 20 \text{ GHz}$$





◦ By Praveen Sah

Thank You for Attending !
Questions please ?

“Creativity is thinking up new things. Innovation is doing new things.” ~ Theodore Levitt